



EDU TERIA

72nd BPSC

Prelims Online Batch

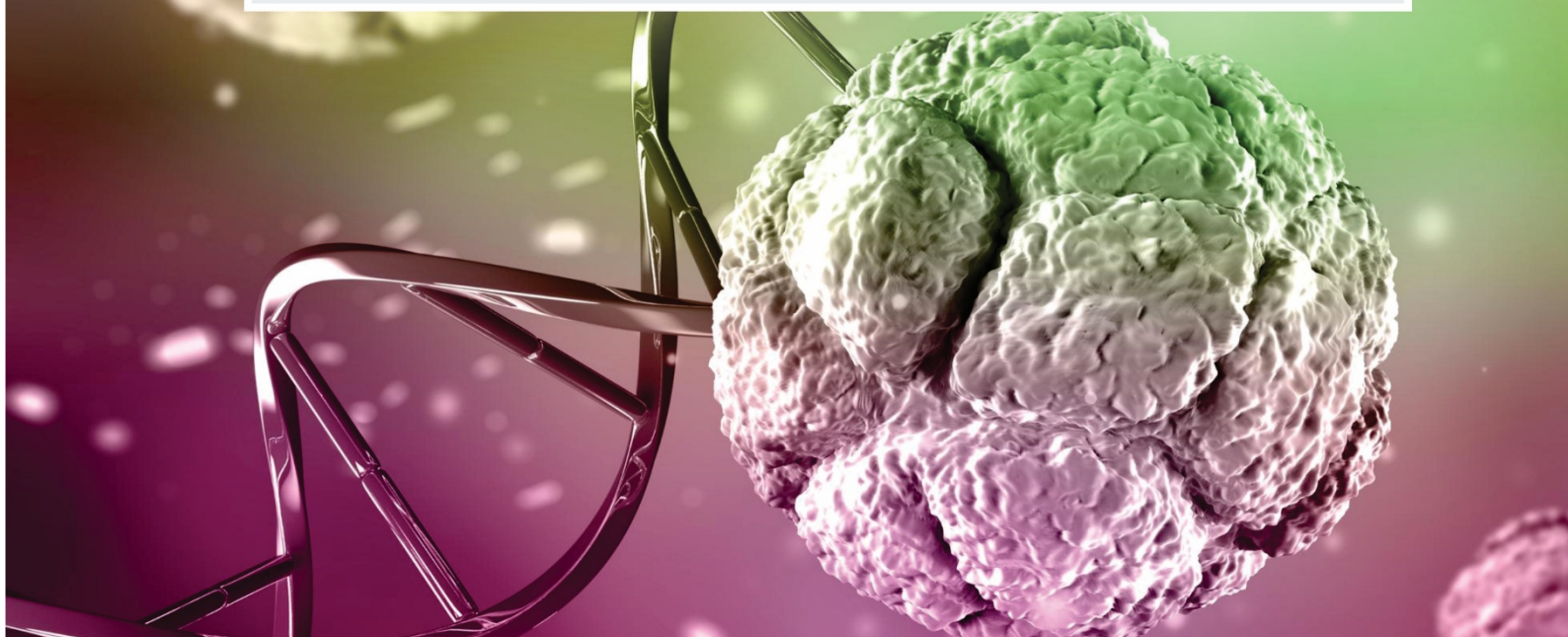
BIOLOGY

Topic wise
Notes

LECTURE

27

**Infectious and Non-Infectious
Diseases - 02**



Infectious and Non-Infectious Diseases - 02

Leprosy/Hansen's Disease

Hansen's disease (leprosy) is a chronic infection caused by *Mycobacterium leprae* bacteria, primarily affecting skin, nerves, and respiratory mucosa.

Pathogen : *Mycobacterium leprae*

Mode of infection

- Airborne
- Droplet infection

Symptoms:

- Discoloured patches on skin without sensation.
- Ulcer and nodule formation in skin and nerves.
- Ulceration and wasting fingers and toes.
- Ulcer and nodule formation in skin, face, earlobes and nerve.
- Loss of eyebrows or eyelashes.

Test for diagnosis

- Skin and nerve biopsy
- Ziehl - Neelsen method test

Treatment:

- MDT (Multi drug therapy)

Vaccine:

- India is first country to develop vaccine against leprosy named as *Mycobacterium indicus pranii* (MIP)

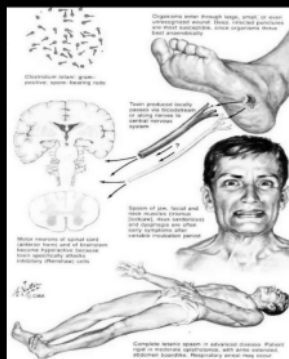
Tetanus / Lock jaw

Pathogen : *Clostridium Tetani*

Mode of infection:

- When any cut or broken skin in foot comes in contact with dirt, poop (feces), or spit (saliva) contaminated with this bacteria.
- Wounds caused by an object puncturing the skin (puncture wounds), like a rusted nail or needle Burn area infection Surgical procedures Dental infections Insect bite.

❖Symptoms:-



- ❖ Lockjaw
- ❖ Painful muscle stiffness all over the body
- ❖ Stretching of Facial muscle
- ❖ Trouble swallowing

Affecting Organ:

- A toxin called Tetanospasmin is secreted by bacteria that inhibit nerve signals from spinal cord to muscles.
- **Test for diagnosis:** Doctors diagnose tetanus based on a physical exam, medical and immunization history, and the signs and symptoms of muscle spasms, stiffness and pain. Laboratory tests generally aren't helpful for diagnosing tetanus

Treatment:

- Immediate treatment with medicine called human tetanus immune globulin (TIG)

Antibiotics

- Drugs to control muscle spasms

Vaccine

- Tetanus toxoid (TT)

Whooping cough / Pertussis

Pathogen : *Bordetella Pertussis*

Mode of infection:

- Airborne
- Droplet infection

Symptoms:

- Fits of many rapid coughs
- followed by a high-pitched "whoop" sound.

Fever

- Fatigue after coughing fits

**Test for diagnosis:**

- Antibody test

Treatment:

- Antibiotics Vaccine
- DPT

Plague:

Pathogen: Yersinia pestis

Mode of infection:

- Spread through Rat flea

Symptoms:

- Fever
- Headache
- Muscle Pain

Treatment

- Antibiotics

Pathogen Virus**Example:**

Disease	Causative Agent
COVID-19	Coronavirus (SARS-CoV-2)
AIDS	HIV (Human Immunodeficiency Virus)
Rabies	Rabies virus
Polio	Poliovirus
Influenza	Influenza Virus
Dengue	Dengue Virus

Polio / Poliomyelitis

Pathogen: Polio Virus

Mode of infection

- Spread through contaminated food and contaminated water (waterborne).

Symptoms

- Fever & Headache Paralysis

Affecting organ

- First intestinal tract
- Soon affected CNS (central nervous system)

Vaccine OPV

- Salk vaccine

Influenza**Pathogen**

- Influenza virus / flu virus

Mode of infection

- Airborne
- Droplet infection

Symptoms:

- Fever, Headache
- Runny nose, cough & cold etc.

Affecting organ:

- Lungs, Throat, Nose, Eyes(conjunctivitis)

Treatment:

- Proper rest
- Antipyretic
- Cough syrup
- Analgesic
- Antiviral medicine

Vaccine:

- live inactivated influenza vaccine Trivalent inactivated influenza vaccine (TIV)

Measles

Pathogen: Morvilli Virus

Mode of infection:

- Airborne
- Droplet infection

Symptoms:

- High grade fever.
- Red spot with white centre inside mouth. (Koplik's spots)
- Red rashes on whole body.

Treatment:

- Increased fluid intake
- Antipyretic

Vaccine:

- MMR(Measles, Mumps, Rubella)

Rubella

Pathogen: Rubella virus

Mode of infection

- Airborne
- Droplet infection

Symptoms

- Mild fever.
- Enlarged lymph nodes near neck and ear
- Red rash beginning on the face which spreads to the rest of the body.

Treatment

- Increased fluid intake
- Antipyretic

Vaccine:

- MMR (Measles, Mumps, Rubella)



**Mumps****Pathogen :** Mumps virus**Mode of infection (संक्रमण का तरीका)**

- Airborne
- Droplet infection

Symptoms

- Mild fever.
- Swelling in parotid gland

Treatment

- Increased fluid intake
- Antipyretic
- Anti-inflammatory drug

Vaccine

- MMR (Measles, Mumps, Rubella)

Chicken Pox**Pathogen :** Varicella-zoster virus (VZV)/ Pox virus**Mode of infection**

- Airborne
- Droplet infection

Symptoms

- Fever
- Dew drop like rashes on whole body

Treatment

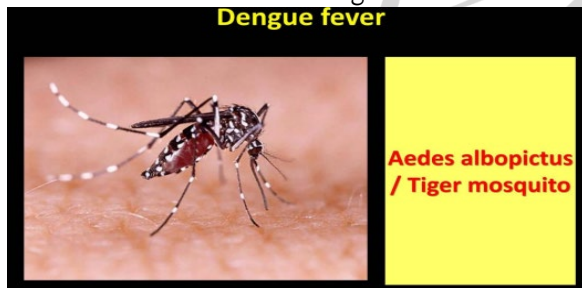
- Analgesic
- Antihistamine

Vaccine

- Varicella vaccine

Dengue fever**Pathogen :** Arbovirus**Mode of infection**

- Vector is Aedes aegypti or can be Aedes albopictus (tiger mosquito), virus is present in saliva of infected mosquito, transmits virus to humans through the bites.

Dengue fever**Symptoms**

- Break bone fever
- Pain in muscle and joint
- Decrease in count of platelets in blood
- In severe cases, there is serious bleeding and shock in patients, which can be life threatening.

Diagnosis & test

- Tourniquet test
- Treatment
- Antipyretic
- Analgesic
- Increased fluid intake
- Platelets transfusion if needed

Vaccine

- Still in trial phase in India.
- Dengvaxia vaccine in USA and some other country.

Chikungunya**Pathogen :** Toga virus**Mode of infection**

- Vector is Aedes albopictus (tiger mosquito) or can be Aedes aegypti, virus is present in saliva of infected mosquito, transmits virus to human through the bites.

Symptoms

- Fever
- Joint pain
- Arthritis

Treatment

- Antipyretic
- Analgesic
- Increased fluid intake

Note: No vaccine**Yellow fever (पीत ज्वर)**

- Yellow fever virus

Mode of infection:

- Vector is mainly Aedes aegypti or can be Aedes albopictus (tiger mosquito), virus is present in saliva of infected mosquito, transmits virus to human through the bites.

Symptoms:

- Fever & headache
- Jaundice
- Vomiting & Nausea
- Muscle pain

Treatment

- Antipyretic
- Analgesic
- Increased fluid intake

Vaccine is Yellow fever vaccine (F-Vax)

AIDS (acquired immune deficiency syndrome)**Mode of infection:**



- Infected fluid (Blood)
- Saliva
- Semen (unprotected sex)
- Mother's milk

Affected part:

- Lymphocyte Cells (WBC)

Protozoa Disease**1. Malaria**

- Causative agent Plasmodium
- **Vector:** Female Anopheles mosquito
- **Symptoms:** Fever with chills, sweating, anemia
- **Transmission:** Mosquito bite

Disease	Causative Agent
Malaria	Plasmodium
Amoebiasis	Entamoeba histolytica
Kala-azar	Leishmania
Sleeping sickness	Trypanosoma
Giardiasis	Giardia lamblia

2. Amoebiasis (Amoebic Dysentery)

- Causative agent Entamoeba histolytica
- **Symptoms:** Diarrhea, abdominal pain, mucus in stool
- **Transmission:** Contaminated food and water

3. Sleeping Sickness

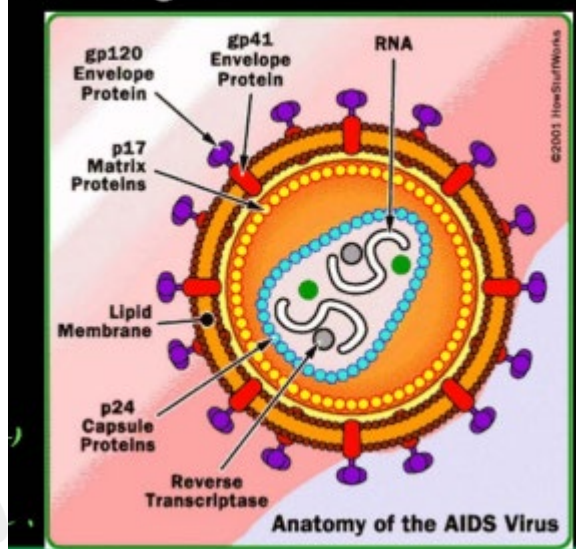
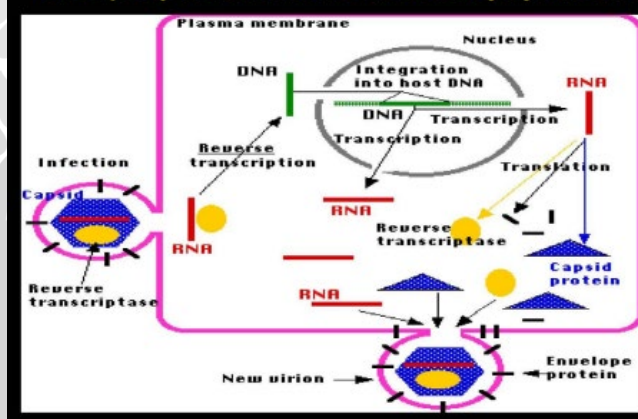
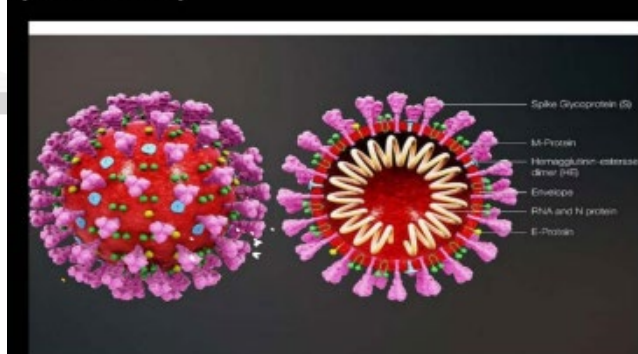
- **Causative agent :** Trypanosoma
- **Vector:** Tsetse fly
- **Symptoms:** Fever, confusion, disturbed sleep cycle

4. Kala-azar Leishmaniasis)

- Causative agent
- **Vector :** Sandfly
- **Symptoms:** Fever, weight loss, enlarged spleen and water

5. Giardiasis

- **Causative event:** Giardia Lamblia
- Transmission Contaminated water

❖ Pathogen :-**AIDS(Acquired immune deficiency syndrome)****Coronavirus Disease 2019 (COVID 19)****❖ Pathogen :-****Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)**

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